



Potential of patent image data as technology intelligence source[☆]

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ABSTRACT

With recent advances in natural language processing and data analytics techniques, useful insights can be extracted not only from bibliographic data but also from the descriptive data of patents. Now, those advances have enabled the use of patent image data as a source of technology intelligence in addition to the two conventional types of patent data. Accordingly, this study focuses on the potential of patent image data and proposes an analysis method for investigating product/service/technology structures using block diagram images among the several types of images in patent documents. Using keywords extracted from patent block diagrams, the following four applications were introduced: (1) analysis of technology evolution, (2) in-depth investigation of technological elements, (3) comparative analysis with competitors, and (4) search for similar patents. The research findings of a case study on mobile earphone technology indicate that keywords are closely related to technological elements, and the four applications are found to be feasible. This study is among the first attempts to support technology intelligence using patent image data. It is also expected to be beneficial in subsequent studies and in practice, wherein patent image data convey valuable information regarding inventions.

1. Introduction

In the rapidly changing external environment, technology forecasting and trend analysis have been regarded as useful mechanisms for identifying new business opportunities. For this purpose, patents have been frequently used. Patents are among the most reliable data sources for technology intelligence (Lee, Kogler & Lee, 2019) and are essential for analyzing technology development (Pilkington, Lee, Chan & Ramakrishna, 2009; Zhang, 2011). In particular, the recent increase in patent applications, together with advances in analytic techniques, has facilitated the use of patent information as a valuable source of technology intelligence. These approaches, which have evolved from the analysis of bibliometric information, including patent index analysis and patent citation analysis to examine text information (e.g., titles, abstracts, claims, or descriptions of patent documents), have been the focus of a recent patent analysis.

The application of text mining to derive technology intelligence has enabled significant advances in patent analysis (Tseng, Lin & Lin, 2007a). Relevant studies have extracted core technological contents from patent documents in the form of keywords or keyword clusters for a wide range of further analyzes (Noh, Jo & Lee, 2015; Yoon, Lee & Lee, 2010). Accordingly, keyword extraction is an essential process of patent analysis, which determines the quality of patent analysis results. Several approaches for keyword selection, including the use of expert domain knowledge (Kim, Suh & Park, 2008; Lee, Kim, Song, Park & Shin, 2014), have been proposed.

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These techniques consider the keyword distribution within patent documents, such as text frequency and term frequency-inverse document frequency (TF-IDF) (Lee, Yoon & Park, 2009; Li, Wang & Hong, 2009; Noh et al., 2015), and customize the process in the selection of the different parts of patent documents (Xie & Miyazaki, 2013).

Although the keyword-extraction process has significantly advanced with the development of natural language processing techniques, current approaches to keyword selection remain limited. First, the general keyword-selection techniques adopted in existing studies (i.e., TF-IDF or text frequency) may not be suitable for identifying high-quality keywords that best describe the characteristics of an invention. Patent documents consist of various types of sentences, including those introducing prior art for an invention or explaining the general characteristics of relevant technological fields. When keywords are selected from those sentences, general approaches to keyword selection may produce rather general information instead of the information that best describes the patent invention (Kim, Park & Lee, 2019). Second, the keywords selected for ex-post analysis must be differentiated from those used for analysis. For example, for investigating technology structure, the keywords should be confined to those representing technological elements. On the other hand, to examine the evolution of technological functions, these terms should focus on technological purposes and effects. The title, abstract, and claims are a mixture of various information, which can make it difficult to extract purpose-specific information, or, at the very least, complicated data preprocessing is required to select only the necessary data. Indeed, to resolve this problem, patent semantic analysis methods such as subject-action-object (Kim et al., 2019) and technology-relation-technology (An, Kim, Mortara & Lee, 2018) have been proposed. Despite that these methods can improve the understanding of keywords, their clarity in presenting the features of patented inventions is insufficient.

Different from existing studies that focus on text data, this study emphasizes the potential of patent image data in keyword selection and analysis. A patent document generally has three types of information: bibliometric, descriptive, and image data. Among these, image data are generated for presenting a complex structure or procedure effectively in a summarized manner. Thus, they are likely to contain only a refined set of keywords for that purpose. Furthermore, image data may be linked to text data within patent documents to support detailed analysis where necessary. Although the derivation of meaningful technology intelligence is expected from patent image data owing to the advances in image analysis techniques, insufficient interest has been given to their potential utilization.

To fill the research gap, this study aims to develop a patent image analysis method for investigating product/service/technology structures using the block diagram of several types of images in patent documents. Patent drawings are of several types: abstract drawing, graph, flowchart, character (symbol), gene sequence, program listing, chemical structure, block diagram, table, and mathematics (Song et al., 2019). Each of these has its own value as a source of technology intelligence. For example, block diagrams present the structure of technical components and their relationships, and flowcharts describe the overall flows of tasks for the patented invention. Hence, block diagrams can be utilized for capturing information about product/service/technology configurations, and flowcharts are for operational procedures. Among those, we selected the block diagram as a target for analysis in this study. Block diagrams are generally used to present the product/service/technology structure to which a patent invention is applied; hence, the keywords in block diagrams tend to indicate the product/service elements. In particular, the elements closely positioned within the diagram are probably more related and more closely interact in the physical system. Thus, the essential keywords for representing the structure of the product/service/technology can be identified from the block diagrams and used for the systematic analysis of evolution in the structure.

In the case of mobile earphones, three steps are required to achieve this objective. First, block diagrams of patent documents from which text information is extracted are collated to construct a keyword set for further analysis. Second, a keyword network that indicates the relationships among keywords is constructed based on the keyword location information in the block diagram. Here, the keywords that are closely located in the diagram are assumed to be more highly correlated. Finally, the keywords extracted from block diagrams are further analyzed in terms of technology structure. This process involves the (1) analysis of technology evolution, (2) in-depth investigation of technological elements, (3) comparative analysis with competitors, and (4) search for similar patents. This study is one of the earliest attempts to support technology intelligence using patent image data. It is anticipated to be useful for future studies and for implementation in practice in order to draw valuable information regarding inventions that are conveyed by patent image data.

The remainder of this paper is organized as follows. Section 2 introduces the existing literature on the use of text mining to derive information from block diagrams in patent documents for patent analysis. Section 3 elaborates on the proposed approach to analyze the information from patent block diagrams. Section 4 presents the validation of the proposed approach based on a case study of mobile earphones. Finally, Section 5 discusses the contributions of this work and future research directions.

2. Literature review

2.1. Patent analysis

Patent analysis can aid in identifying and understanding key technological fields by counting and comparing the cumulative numbers of patents across all technological fields and comparing them (Wartburgh, Teichert & Rost, 2005). A wide range of patent analysis methods have been recently proposed because of the enhanced accessibility of patent databases along with advances in data analytics. These analysis techniques can be mainly classified into two categories (i.e., *bibliometric* and *text analyzes*), according to their target, using structured and unstructured patent information, respectively.

Patent bibliometric analysis utilizes a large amount of patent bibliometric information, such as assignees, assignee countries, application dates, registration dates, and citations to support strategic decision-making on technology and innovation

(Daim, Rueda, Martin & Gerdtsri, 2006; Lai & Wu, 2005). Given the large quantity of the relatively structured form of data, several tools have been developed to process them (Porter, 2005; Pilkington, 2003; Porter, Watts & Anderson, 2003). In particular, patent citation analysis has long been used to examine knowledge flows, sometimes being used together with publications data (e.g., Huang, Yang & Chen, 2015).

In contrast, patent text analysis relies on text-mining techniques to identify meaningful implications from the descriptions of invention. Relevant studies can be divided into several research streams. The first stream concentrates on the application of text-mining techniques customized to patent documents. For example, this stream includes the investigation of the best methods for extracting the optimal set of keywords that can explain the patented invention (Li et al., 2009; Noh et al., 2015), deriving meaningful concepts from patent documents (Choi, Kim, Yoon, Kim & Lee, 2013, 2012; Yoon, Park & Kim, 2013; Choi & Hwang, 2014), selecting appropriate keywords for patent search (Xie & Miyazaki, 2013), or developing improved patent similarity measures (An, Li, Xu, Chen & Sun, 2021). The second stream focuses on improving the efficiency of patent-related tasks using keywords extracted from patent documents. One of the most representative examples is the application of text mining to automatically cluster patent documents (e.g., Lai & Wu, 2005), assign patent classification codes to new patent applications (Kim & Choi, 2007), or generate patent summaries (e.g., Codina-Filbà et al., 2017). This can reduce the enormous amounts of human resources required to classify patent documents (Chakrabarti, Dom, Agrawal & Raghavan, 1998b; Fall, Torcsvari, Benzineb & Karetka, 2003; Lee et al., 2009; Liang, Tan & Ma, 2008; Tseng et al., 2007a). The final research stream is centered on deriving valuable technology intelligence from patent information, where the extracted patent keywords are employed to investigate technological trends and capture new technology opportunities (Yoon & Park, 2004; Kim et al., 2008; Kim, Gazzola, Lee, Kim & Jeong, 2014; Li et al., 2009; Son, Suh, Jeon & Park, 2012; Tseng, Wang, Juang, & Lin, 2005).

Although numerous studies have been implemented on patent analysis, few attempts have been made to utilize patent image data as a source of technology intelligence. Various algorithms are currently available for analyzing image data; hence, the investigation of the potential of patent image data is feasible. Ultimately, exploring various means of creating value from image analysis is worth considering because the derived information can complement bibliometric and text data to understand technology trends and identify new technology opportunities.

2.2. Patent drawings

Patent documents mainly consist of four parts: abstract, claims, description, and drawings. The first three are in text form, whereas drawings are in image form. Such drawings are included in patent applications to aid in explaining the invention through the summary of technical contents from a particular perspective. In view of this, patent drawings are likely to contain valuable information on the key features of a patented invention. Some recent studies have highlighted the importance of patent drawings. However, most have concentrated on retrieving similar patents based on the images (Wu, Lam, Mehtre & Gao, 1996; Eakins, 2001; Tiwari & Bansal, 2004; Stefanos et al., 2010; Stefanos, Anastasia & Ioannis, 2012) and supporting experts in efficiently searching for patents (Stefanos et al., 2010). In contrast, only a few studies have analyzed the content of patent drawings for driving technology intelligence. Because the figures and diagrams in patent drawings are effective for describing an invention, as well as emphasizing the key features mentioned in the abstract, claim, or description in an easy-to-understand form, they deserve consideration as they can provide valuable information.

Patent drawings can take several forms, each of which has its own value as a source of technology intelligence. Table 1 summarized the characteristics of various types of patent drawings, as well as their value as a source of technology intelligence, based on the previous studies (Vrochidis et al., 2010; Hanbury, Bhatti, Lupu & Mörzinger, 2011; Csúrka, 2017).

Among those drawings, this study is focused on block diagrams. A block diagram is a drawing describing the relationships among the various parts of systems (or machines) using squares or rectangles connected to each other by straight lines. Because text describing the elements of technologies (i.e., parts of systems) fills these shapes, the relationships among texts can also indicate the relationships among the elements. Most studies on the design of keyword-based patent analysis assume that the extracted keywords for in-depth analysis sufficiently represent the contents of patent documents. Although a few studies have suggested keyword-selection approaches customized for patent analysis (Xie & Miyazaki, 2013), most lack the keyword-validation process. In this work, a set of keywords describing the features of technological structures are expected to be obtained by referencing the texts in block diagrams. This can reduce noisy information found in other parts of the patent document and enables useful analyzes in terms of product/service/technology structure.

3. Research framework

3.1. Overall research process

Fig. 1 illustrates the overall research process. First, patent image data from which blocks with text are recognized are collected. Keywords from the blocks are selected for further analysis, and their relationship matrix is established. Finally, four types of technology intelligence are derived based on the matrix.

3.2. Stage 1: patent image data collection

First, representative drawings, which are first set as drawings in the Google Patent site, are collected. Then, only those with the term “block diagram” in their titles are adopted for analysis (Fig. 2).

Table 1
Patent drawings in the patent document.

Type	Description	Exemplary intelligence
Abstract (Technical) drawing	A drawing of the shape, structure, size, and processing method of the invention accurately, simply, and clearly by using lines, characters, and symbols to convey the inventor's requirements	Shape, structure, size and processing
Character & symbol	Characters or symbols that exist in the patent image	Technological elements
Chemical structure	Atoms that make up a compound, including those that show the mutual relationship by the chemical reaction of the constituent atoms	Structure of atoms and their relationships
Block diagram	A visualization of technical information by drawing, and symbolizing it with a two-dimensional geometric mode	Structure of technological elements and their relationships
Flowchart	A type of diagram that shows a workflow or process using different types of boxes, followed by arrows, to show a solution model for a particular problem	Process of actions
Gene sequence (DNA)	An image of the arrangement of the letters of the four chemical substances A (adenine), T (thymine), G (guanine), and C (cytosine) that make up DNA	Structure of technological elements and their relationships
Graph	A structure in which pairs of some objects form a set of related objects	Performance
Mathematics & table	Notation and symbols used to indicate mathematical relationships, and mathematical concepts being organized and expressed in formulas	Technological elements and their relationships
Program listing	Source code in which the invention algorithm is implemented	Process of actions
Photo	Black-and-white or color photographs that exist in the patent document	Product/service/technology designs

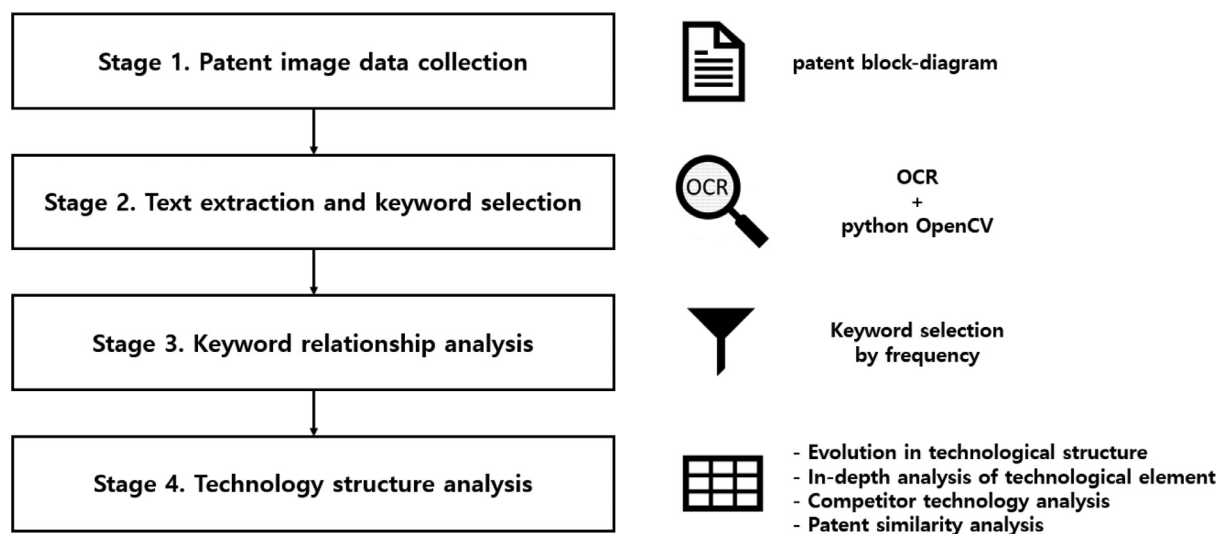


Fig. 1. Overall research process.

The purpose of this study is to demonstrate the potential value of patent drawings as a source of technology intelligence, where their possible applications may vary depending on their type. For example, the block diagrams are useful for investigating technology trends in terms of product structure. However, not all patents contain information about product structure; some patents are for a specific product component, service, or function. Consequently, only a portion of patents are available for analyzing technological trends from the product structure perspective. On the other hand, other patent drawings such as flowcharts or chemical structures can have other applications. For example, flowcharts are useful in investigating operational procedures, and chemical structures are available for identifying drug repositioning opportunities. Likewise, not all types of drawings are included in a single patent; each type of drawing has its own particular information value.

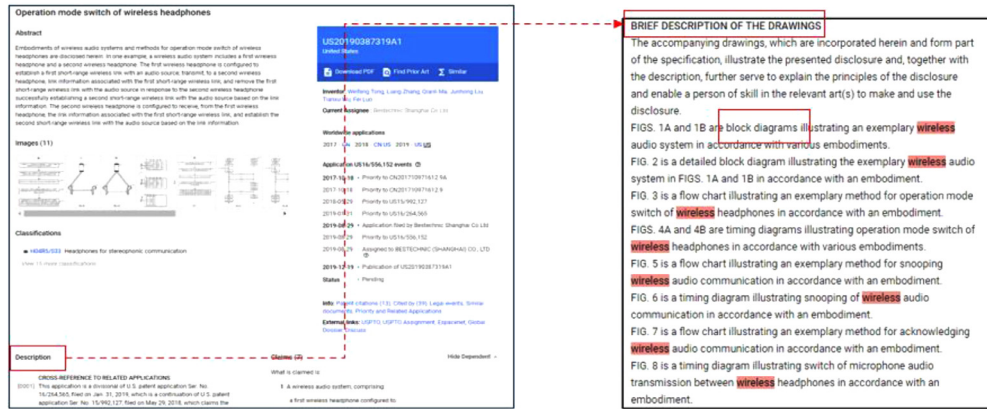


Fig. 2. Patent image data.

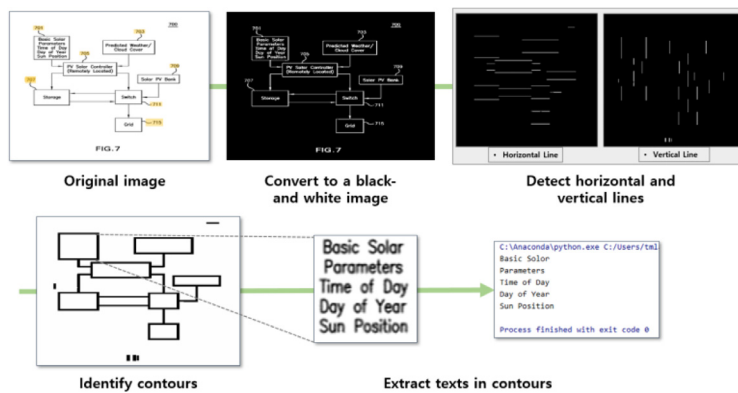


Fig. 3. Text extraction process.

3.3. Stage 2: text extraction and keyword selection

At this stage, texts are extracted from the patent images (Fig. 3). The horizontal and vertical lines in the target image are detected, based on which contours (rectangles) are identified using the Python OpenCV library and findContours function. The positions of contours containing text are defined as center coordinate values. Finally, the text is identified using an optical character recognition (OCR) engine provided by the Python-pytesseract library. This library is based on Google's Tesseract 4.0, which is regarded as one of the most widely adopted (performance rate of approximately 90%) (Satyawati et al., 2019; Sporic, Cuşnir & Boiangiu, 2020). The poorly recognized keywords need to be removed before further analysis is carried out. Thus, results that are more reliable can be obtained by introducing more advance open sources or APIs.

After text recognition, we selected keywords for further analysis based on their characteristics (Yoon et al., 2010). First, stopwords and general terms need to be removed. To do this, we adopted 179 English stopwords from the Python's NLTK library, one of the most frequently used ones for natural language processing. We also introduced the Oxford 3000 Dictionary to remove general words. This dictionary contains a list of words that appear most frequently in English text based on the British National Corpus and the Oxford Corpus Collection. This text preprocessing helps us identify only the keywords worth investigating and generate meaningful implications from further analysis based on the keywords. Then, with the remaining keywords, we carried out keyword selection. Between the two common approaches for keyword selection (i.e., frequency-based and TF-IDF-based) (Lee et al., 2009; Li et al., 2009), we adopted the frequency-based approach in this study. We assumed most of the terms in the drawings were significant given that the patent attorney or inventor implemented the keyword-screening process when creating the drawings.

3.4. Stage 3: keyword relationship analysis

In this study, we assume that the distance between the texts indicates the degree of relationships among them. For example, if a pair of keywords are observed frequently in the same patent drawings, they are likely related to each other. Moreover, if keywords are close to each other in the patent drawings, they are likely closely linked to each other. Indeed, two technological elements (in the form of blocks containing a keyword) are next to each other when they need to attach to each other, with one affecting the other, or need to be integrated into the same application. Accordingly, in this study, we propose an approach for analyzing texts in

Table 2
Binary patent-to-keyword matrix example.

Patent	X_axis	Y_axis	A	B	C	D	E	F	...
US1	1	1	1	0	1	0	0	0	...
US1	251	7799	0	0	1	1	1	0	...
US1	321	5541	0	0	0	1	0	0	...
US1	215	2215	0	0	0	0	0	1	...
US1	45	664	0	1	0	1	0	1	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

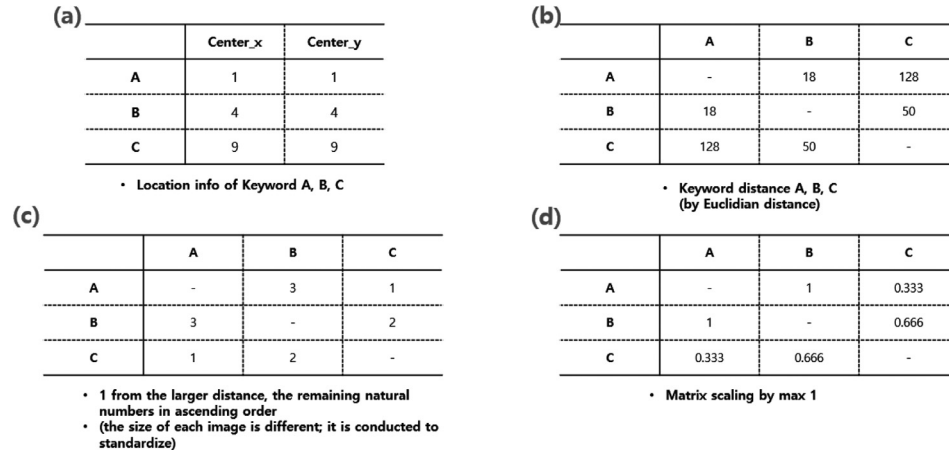


Fig. 4. Keyword-to-keyword matrix example.

block diagrams to investigate the technological elements on the premise that the actual distance between keywords in patent block diagrams can be a proxy for the relationships between them.

To measure the distance between keywords, first, a binary patent-to-keyword (P2K) matrix is created using the set of keywords selected in the previous stage (Table 2), where information regarding the position of the contour and its corresponding keywords is provided. Hence, this matrix has keywords as columns and patent IDs with contour positions as rows. The cell value is 1 if a keyword (j) exists in the contour (X_axis, Y_axis) of Patent P(i); otherwise, it is 0, where the X_axis and Y_axis indicate the keyword locations in each patent drawing.

The following step obtains a keyword-to-keyword (K2K) matrix based on the location of keywords derived from patent images using the P2K matrix. Assume a matrix with three keywords (A, B, and C) for patent P(i) (Fig. 4(a)). The Euclidean distance between two keywords can be calculated using the keyword location information (Fig. 4(b)). Then, the matrix values are transformed by assigning ranks by distance in ascending order (Fig. 4(c)). Finally, the matrix values are normalized by dividing the cell values by the maximum values; this scales the matrix to values in the range of 0–1 (Fig. 4(d)). This normalization process is necessary because the number of keywords in images and the sizes of images differ. Accordingly, the K2K matrix for P(i) is created for all patents, and then the global K2K matrix is developed using the arithmetic mean values of keywords (j1, j2) for all relevant K2K matrices.

3.5. Stage 4: technology structure analysis

In this stage, an integrated K2K matrix (P) is visualized in the form of a network using the Gephi program; the nodes represent keywords, and each edge represents the distance between two keywords. Thus, this network is a technology structure map in which technological elements that are physically or functionally associated are positioned close to each other in the patent drawings; moreover, the elements are more likely to have strong connections. Then, a modularity analysis is performed to cluster the closely related elements systematically. Modularity is a method for measuring the network structure; hence, it can be applied to divide the technological structure into several groups of elements.

Once the technology structure map is established, various follow-up analyzes become feasible using the map to improve the understanding of the technology and support strategic decision-making. Four technology intelligence analyzes are suggested. The first is the analysis of evolution in the technological structure. The evolution of a product or service system can be investigated by the changes in keywords and the relationships among keywords over time. The second is the in-depth analysis of technological elements in relation to other elements. The importance of a keyword and its relationships with other keywords can be examined over time by focusing on a single keyword rather than a network; this enhances keyword understanding. The third is a competitor technology analysis with respect to the technological structure. If a company compares its own technology (product or service) with competitors, it can easily comprehend its distinguishing features from the perspective of technological structure. The last is patent

patent_figure_num	text	center_x	center_y
US-2010023320-A1.jpg	Multimodel Device	381	290
US-2010023320-A1.jpg	Multimodel Device	381	290
US-2010023320-A1.jpg	Context Description Grammar	2383.5	402
US-2010023320-A1.jpg	Text to Speech Engine	1383	417
US-2010023320-A1.jpg	Trans Speech ceiver Coder	1379.5	663
US-2010023320-A1.jpg	Speech Coder	1498.5	662.5
US-2010023320-A1.jpg	Trans receiver	1260.5	662.5
US-2010023320-A1.jpg	Speech / Recognition Engine	1373	909.5
US-2010023320-A1.jpg	Parser	1373	1131.5
US-2010023320-A1.jpg	Update Manager	2386	1136.5
US-2010023320-A1.jpg	Network Interface	1373	1375.5
US-2010023320-A1.jpg	NonSpeech Interface	1373	1610
US-2010023682-A1.jpg	HOST MOTHERBOARD 419	1101.5	159.5
US-2010023682-A1.jpg	UPSTREAM INTERFACE 34	970	647.5
US-2010023682-A1.jpg	VIRTUAL BUFFER BRIDGE 32	716.5	994
US-2010023682-A1.jpg	VIRTUAL STORAGE PROCESR	1701.5	994
US-2010023682-A1.jpg	VIRTUAL BUFFER BRIDGE 32	810.5	994
US-2010023682-A1.jpg	SMART STORAGE TRANSACTION MANU	1261	994.5
US-2010023682-A1.jpg	VIRTUAL STORAGE BRIDGE 43	1289	1416
US-2010023682-A1.jpg	VIRTUAL STORAGE BRIDGE 42	801	1416
US-2010023682-A1.jpg	MAPPER NVM 46 CTRL 7p	1410.5	1741.5
US-2010023682-A1.jpg	NVM CTRL	792	1741.5

Text by each contours

patent_figure_num	text
US-2010023320-A1	Multimodel Device. Multimodel Device. Context Description Grammar. Text to Speech Engine. Trans Speech ceiver Co
US-2010023682-A1	HOST MOTHERBOARD 419. UPSTREAM INTERFACE 34. VIRTUAL BUFFER BRIDGE 32. VIRTUAL STORAGE PROCESR. VIRTU
US-2010023858-A1	POWER SUPPLY UNH. BROADCAST RECEIVING MODULE. AUDIO OUTPUT MODULE. MOBILE COMMUNICATION MODU
US-2010060586-A1	lgsModules E Weather Widget e Stocks Widget Calculator Widget Alarm Clock Widget Dictionary Widget S. B cuaw
US-2010060664-A1	Wireless controller. Display. Short range Wireless Interface. gbgz/ High speed serial link. Touchscreen controller. Display co
US-2010078471-A1	7Device 1 (Payee). Device 1 (Payee). Financial Server(s). Device 2 (Payor).
US-2010078472-A1	Video Controller. NFC Device. 7 a 4 U 5 = 4 mov =+ moa slim r L 4 at. U a l a U a = l "T" roa == are ma " = ma. Comm
US-2010081487-A1	RF TRANSCIEVER 110. ECHO/NOISE CANCELLATION 180.
US-2010082490-A1	. Touch Screen. Accelerometers. User Input Device. Network Device. Storage Memory. Video Processing Circuitry. Position
US-2010090656-A1	26, 28 Power Regulation and/or Power - Storage Circuit. RF Power Harvesting. Power Storage (e.g. Battery, Capacitor, etc.)
US-2010097250-A1	SOURCE DIGITAL SYSTEM 104 106 108 vbeo vbeo CAPTURE ENCODER TRANSMITTER. TRANSMITTER. VIDEO ENCODER.
US-2010118819-A1	EGRESS PACKET QUEUE ULU * J. INGRESS PACKET QUEUE DO ood.
US-2010120470-A1	POWER SUPPLY UNIT. I AN INPUT UNIT CAMERA MICROPHONE 17 0US2010120470A1.jpg7911371301176MULTIMEDIA N
US-2010121156-A1	AGE I IFR. CHANCE SAE. ICK RANGE. NEDICA NSVRANCE Bots. NEDICA NSVRANCE (NA. CORBUKCK amim. CORBUKCK CA 4
US-2010130137-A1	TRANSCIEVER TRANSCIEVER. eNB 817 TRANSCIEVER PROCESSOR MEMORY TRANSCIEVER 815. MME 333 PROCESSOR. TR
US-2010138680-A1	PRESSENCE DETECTOR. SENSING POINTS. IN/OUT OF HAND DETECTOR. DEVICE APPLICATIONS). I/O SELECTOR. SPEAKER
US-2010145861-A1	324 T. PALM 322 350 324 WEB SERVICES ACCOUNT SERVICES PAYMENT SERVICE 336.
US-2010161658-A1	Selected: Pub. Find around me: pubg within 2 km. orner Pub Click for special offer IOO (O. Corner Pub Click for: spe
US-2010175014-A1	AUTO DIRECTOR. Rear Console. #NAME?. Multi Function. Center Console. Rear Console. Instrument Panel. Control Popel
US-2010180063-A1	vr 2 Ao hee el ced 2 es I.A.
US-2010188304-A1	Navigation Reference System. Antenna Array. Network Management Server. Antenna Control Unit. Gimbal Assembly. . D a
US-2010194857-A1	POWER. RADIO TRANSMITTER. IMAGE OUTPUT. DISPLAY / PROJECTOR.
US-2010201312-A1	COIL. POWER SOURCE.
US-2010210208-A1	MATCHING NETWORK3 136. . MATCHING NETWORK1 132. W MATCHING NETWORK2 134.

Text by each patents

Fig. 5. Text extraction results.

Table 3

Total number of patents for analysis.

	2010–2011	2012–2013	2014–2015	2016–2017
Number of patents initially collected	216	169	179	110
Number of patents with no text detected	2	10	13	9
Number of patents finally used for analysis	214	159	166	101

similarity analysis in terms of the technological structure. The use of text in images, together with conventional search methods, can improve the reliability of patent searches.

4. Case study

4.1. Background

A case study on mobile earphone technology was conducted. With the commercialization of various smart devices and applications, mobile earphone technology has gained increasing interest in the fourth industrial evolution era. As companies endeavor to accelerate the development of mobile earphone technologies to dominate the market, technologies with diverse purposes and effects have also been proposed. In addition, mobile earphones are designed to communicate with other devices; thus, patents in this field are expected to have numerous drawings that show links among the technological elements. In the current case study, the evolution of mobile earphones through the interactions of several converging technologies is examined using patent image analysis.

4.2. Stage 1: patent image data collection

We collected 12,789 patents on mobile earphone technologies published after 2010. The data were obtained from Google Patents using the search query “TAF = wireless earphone and ad >2010.” To investigate key trends, 500 of the most cited patents each year were collected; from these, only those that used block diagrams as representative drawings were selected for further analysis. The patents were grouped into four periods for time series analysis: P1(2010,2011), P2(2012, 2013), P3(2014, 2015), and P4(2016, 2017).

4.3. Stage 2: text extraction and keyword selection

4.3.1. Text extraction and keyword selection results

All texts were extracted from the patent images of collected patents (Fig. 5). Images not containing any texts were eliminated from the database. Table 3 lists the total number of patents for the analysis. As we focused only on the representative images in the form of block diagrams, approximately 32% of 500 patents (160 patents) were available in our analysis. However, the purpose of the analysis is to identify technical keywords and investigate the features of technological elements with a focus on their structure. Given that some patents are for a specific product component, service, or function, not all patents have information about the structure of technological elements. Thus, such limited data can also create a value to understand the technology trends in terms of structure. The analysis can be extended to other non-representative block diagrams, and block diagrams with little full-text description are available for analysis as well.

Once the texts were extracted, those with more than four frequencies were selected, producing a list of 542 keywords: more valuable keywords were produced by the frequency-based selection than the TF-IDF-based selection (Appendix 1 for details). Then, less informative keywords (those in the categories “general” and “others” in Appendix 2) were deleted; 486 keywords remained. In

Table 4
Topic-document distribution based JSD matrix by source in patents.

	Title	Abstract	Claims	Description	Image
Title	0	0.014	0.039	0.032	0.033
Abstract	–	0	0.018	0.015	0.028
Claims	–	–	0	0.014	0.037
Description	–	–	–	0	0.030
Image	–	–	–	–	0

Table 5
Comparison between extracted abstract and image keywords.

Distinguishing keywords Abstracts (254)	Images (260)
Method, plurality, embodiment, invention, request, present, multiple, use, current, various, command, value, product, region, aspect, interaction, technique, item, particular, condition, reference, portion, computing, performance, part, different, field, medical, instruction, identifier, ambient, view	gps, engine, encoder, decoder, cpu, dsp, generator, graphic, baseband, calculator, search, i/o, subsystem, wifi, adc, usb, accelerometer, web, guest, ram, track, capture, definition, modulator, volatile, modem, lcd, shortrange, utility, codec, lan, keypad, rom, dvd, param, tx, rx, dac, pc, msc, timer, mam, a/d, sgsn/pdsn, amp, ui, a/v, rlc, queue, ethernet, vf, stereo, ion, d/a, mp3, earcon, dram, dc/dc, adapter, drx, haptic, mc, sdram, rfid, pdm, pas ...
Common keywords (282) Module, device, interface, unit, display, controller, processor, audio, system, user, memory, data, sensor, power, input, server, communication, application, wireless, service, storage, component, network, video, processing, transceiver, output, widget, content, source, mobile, information, camera, management, circuitry, signal, location, player, image, client, manager, receiver, detector, radio, database, filter, channel, transmitter, microphone, speaker, function, provider, rf, external, circuit, access, group, terminal, phone, driver, battery, program, optical, host, port, computer, speech, peripheral, supply, digital, payment, software, endpoint, center, operating...	

this study, the unit of analysis is a single word rather than a term. Although a single word fails to express rich information, it is the term with the smallest granularity for semantic analysis. In addition, relatively, a robust set of words can be identified with a single-word approach, but terms identified from patent drawings may differ by term-extraction processes and algorithms. Compared to texts in patent abstracts, descriptions, and claims, which are complex in their sentence structures, texts in patent images are short and simple and are therefore terms that are easy to identify.

4.3.2. Validation of the keywords

To prove the value of information presented in patent block diagrams, we first analyzed the differences in the information provided by the five different sources: title, abstract, claims, description, and images (block diagrams). For this purpose, we conducted LDA on the texts from the five sources, after which we compared their topic-document probability distributions using KL-divergence-based Jensen Shannon divergence (JSD). The JSD values range from 0 to 1, where the closer the value to 0 is, the more similar the two probability distributions are. As Table 4 shows, the JSD values between title, abstract, claims, and description range from 0.014 to 0.039, and the JSD values between images and the others – title, abstract, claims, and description – range from 0.028 to 0.037. The values are relatively low because all the texts are used to describe the same invention regardless of their positions in patent documents. However, the comparative analysis offered meaningful implications. The table indicates that the JSD values between title, abstract, claims, and description were less than 0.018 except in one case (the value for titles and claims) whereas all the JSD values for images were greater than 0.028. This result indicates that images seem to have heterogeneous probability distributions, compared to all the other sources.

Second, to investigate such a difference in detail, we focused on the keywords from abstracts and images given the relatively high similarity among titles, abstracts, claims, and descriptions. Accordingly, we extracted keywords from abstracts and images to identify those observed only one of them and both of them as shown in Table 5 (Appendix 3 for details). The analysis results indicate that while general keywords are observed in abstracts (e.g., method, embodiment, invention, and technique), more specific keywords in relation to physical and functional elements are observed in images (e.g., encoder, decoder, CPU, DSP, and generator), supporting the assumption made in this study. The keywords observed only in images are closely related with the components of products/services given the nature of the diagram, pointing out the potential value of patent image data as a source of technology intelligence.

Finally, we examined whether the keywords observed from the images can also be identified easily from the other parts of patent documents. Because the keywords in the images are those already chosen carefully by the inventor (or patent attorney) to summarize their patented invention, particularly those from representative drawings, those observed frequently in the images can be regarded as important. On the other hand, the keywords frequently observed in other parts of patent documents are likely to include numerous noisy (or less important) keywords given that those parts consist of sentences and paragraphs; this makes users apply TF-IDF (term frequency-inversed document frequency) instead of TF to patent descriptions. Thus, we tested the degree to which the keywords observed in images (top 10% keywords sorted by TF) could be extracted from titles, abstracts, claims, and descriptions when TF-IDF and TF were applied to them. Table 6 summarizes the average ranks for those 75 keywords from the list of keywords identified from the other parts of patents, sorted by TF-IDF and TF values, which indicates the 75 keywords are not easily detected unless a large

Table 6

The number of keywords required to extract Top-N keywords in the image (TF-IDF).

Term	Title	Abstract	Claim	Description
module	271.5	64.5	214	411
device	9	31	21	16
interface	62	313	588	615
display	6	126	124	236
processor	193.5	230	607	708
user	14	82	109	40
audio	13	79	80	194
controller	193.5	471	383	709
system	23	442	209	74
server	–	322	444	318
memory	–	203	1134	955
data	193.5	550	741	888
input	105	443	422	533
power	1.5	64.5	170	348
control	21	293	372	330
communication	24	234	172	222
application	66.5	178	164	274
wireless	15	163	260	347
storage	95.5	316	879	1131
sensor	42	125	270	296
transceiver	271.5	387	499	1512
network	12	288	457	275
component	271.5	183	239	651
video	66.5	223	240	423
media	–	524.5	1224	2085
...	...	...	...	...
Average	107.6	303.7	489.7	782.7

Table 7

Percentage of keyword detection in images according to the top-N based on TF-IDF.

Top-N	Title	Abstract	Claim	Description
75	38.7%	14.7%	6.7%	4.0%
150	56.0%	29.3%	17.3%	6.7%
225	66.7%	49.3%	28.0%	14.7%
300	78.7%	61.3%	40.0%	25.3%
375	78.7%	72.0%	45.3%	36.0%
450	78.7%	78.7%	52.0%	42.7%
525	78.7%	84.0%	61.3%	44.0%
600	78.7%	88.0%	70.7%	48.0%
675	78.7%	93.3%	73.3%	53.3%
750	78.7%	94.7%	78.7%	62.7%
825	78.7%	94.7%	82.7%	64.0%
900	78.7%	96.0%	85.3%	72.0%
975	78.7%	96.0%	86.7%	73.3%
1050	78.7%	97.3%	93.3%	73.3%
1125	78.7%	97.3%	93.3%	77.3%
1200	78.7%	98.7%	94.7%	80.0%
1275	78.7%	98.7%	97.3%	81.3%
1350	78.7%	98.7%	98.7%	81.3%
1425	78.7%	98.7%	98.7%	85.3%
1500	78.7%	98.7%	98.7%	85.3%

number of keywords are adopted for analysis (107.6 terms for titles, 303.7 terms for abstracts, 489 terms for claims, and 782 terms for descriptions). Furthermore, only 78% of terms identified in patent images are presented in titles, though a relatively small number of keywords are required to identify all 75 patents.

Table 7 summarizes the ratio of keywords identified from each source of patent documents when top-N terms were selected by TF-IDF. In case of titles, more than 300 terms need to be selected to include 70% of target keywords (75 keywords identified from images). In case of abstracts, 375 keywords are required to detect 75% of target keywords, while 675 keywords are needed for 90% of target keywords. Even more keywords are required in cases of claims and descriptions.

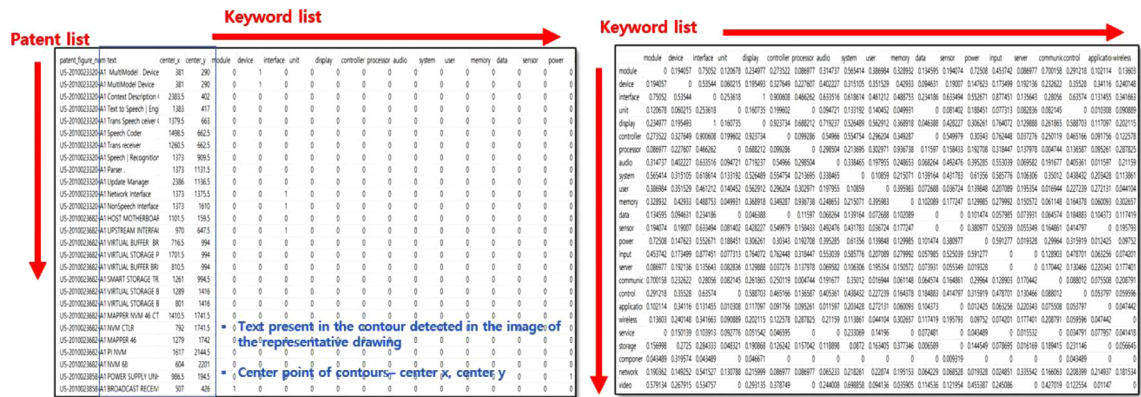


Fig. 6. Keyword relationship analysis results.

Table 8
Modularity analysis results.

Type	Module 1 (28)	Module 2 (85)	Module 3 (22)	Module 4 (20)
Keywords	interface, display, controller, audio, sensor, input, control, output, speaker, digital, cpu, converter, sound, touch, screen, switch, ram, volatile, noise, rom, amplifier, object, usb, lcd, virtual, dac, adc, level	device, processor, user, memory, data, server, application, service, storage, component, network, processing, transceiver, content, source, signal, manager, receiver, detector, radio, database, filter, transmitter, provider, circuit, access, terminal, phone, engine, battery, program, decoder, software, center, voice, logic, generator, cellular, key, ...	module, unit, communication, wireless, mobile, information, location, microphone, supply, broadcast, receiving, area, multimedia, internet, position, shorrange, pressure, local, determination, short	system, power, video, widget, camera, management, circuitry, player, image, client, gps, rf, external, port, operating, proximity, telephone, search, alarm, music, messaging, browser
Label	Interface-related	Device-related	Module-related	System-and power-related

4.4. Stage 3: keyword relationship analysis

Using the final keyword set, a binary P2K matrix was constructed: If a keyword (j) identified from a patent (Pi) appeared in a contour, then the relevant cell assumed the value of 1; otherwise, it was 0. Then, a K2K matrix was developed based on the average of the normalized distances among keywords (j1, j2) for all patents with two keywords (Fig. 6). The size and number of drawings/boxes are different for each patent. Thus, we adopted data concerning only representative drawings, and the distance between keywords were standardized according to the size of drawing. Accordingly, we believe that the global relationships between keywords converge and are meaningful (see Appendix 4 for details).

To investigate the annual trends of and the relationships among keywords, 155 keywords observed in all four periods (P1–P4) were adopted. Thereafter, the relationships among the keywords were analyzed using the Gephi program based on the K2K matrix. The modularity analysis of the network generated four areas (Table 8); that is, the mobile earphone technology can be divided into interface, device, module, and system-and-power areas.

4.5. Stage 4: technology structure analysis

4.5.1. Evolution in technological structure

An edge cut-off value of 1.0 was applied to explore the trends in the technological structure over time in terms of network and keyword features. Fig. 7 presents the evolution of the network structure, where the size of the node and the width of the edge indicate the number of keywords and the degree of relationships among the keywords, respectively. In the first period, various technologies were developed. No distinct technologies were observed although most were in the system-and-power-related area. However, the development effect in system-and-power technologies weakened in the succeeding periods. In contrast, an intensive development in display and device technologies was observed in P2, which intensified in the subsequent periods. The network cohesion also seemed to have been reduced from P1 to P4, which meant the technologies were specialized and differentiated.

Second, technology evolution was examined in terms of unique keywords in each period (Table 5). The emergence of new technologies was identified by new keywords (i.e., keywords other than the 155 common keywords over the four periods). An emergence

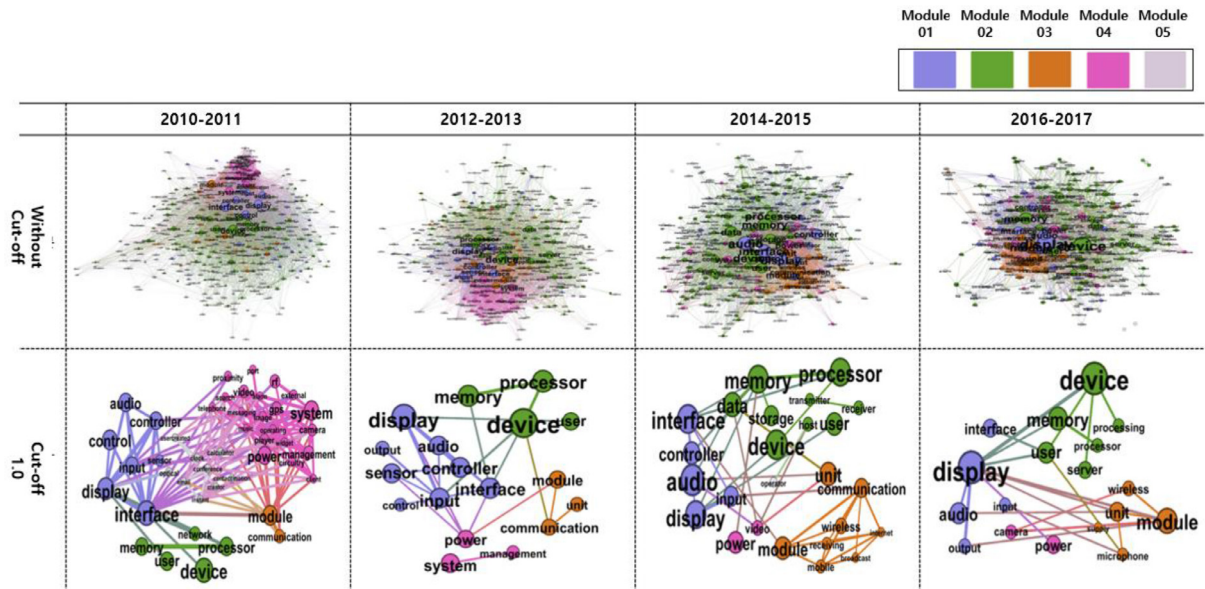


Fig. 7. Network visualization results.

Table 9
Keyword analysis results.

	2010–2011 (P1)	2012–2013 (P2)	2014–2015 (P3)	2016–2017 (P4)
Patents	214	159	166	101
Emerging keywords	215	186	181	117
Emergence value	1.00	1.16	1.12	1.15
Top 20 keywords by frequency	interface, device, display, module, control, system, audio, user, processor, power, controller, memory, input, transceiver, network, wireless, storage, communication, rf, management	device, display, processor, memory, interface, server, input, controller, sensor, system, data, audio, power, module, user, application, communication, processing, storage, unit	audio, device, interface, display, processor, memory, data, unit, controller, control, module, user, power, storage, input, communication, server, system, content, wireless	device, display, module, memory, user, audio, unit, interface, power, server, controller, data, system, wireless, output, communication, camera, processing, processor, source
Unique keywords (number)	internal, exercise, config, trainer, device/global, roller, handover, pad, coder, common, earcon, settlement, loudspeaker, jack, stack, assist, section, purchase, entity, definition, feature, trace, dynamic, po, gadget, servicer, gw, mod (28)	usage, cation, snet, arm, vector, environment, profiler, porosity, zone, zip, frame, active, semantic, credential, whiteboard (15)	endpoint, gain, mike, fingerprint, privacy, protection, sensory, stimulation, workstation, row, track, subband, ad, cookie, pdm (16)	commun, priority, hid, apparatus, companion, preference, emotion (7)

index was designed to measure the degree of emergence.

$$Em(P_i) = N_{ek}(P_i) / N_t(P_i),$$

$N_{ek}(P_i)$: the number of emerging keywords at P_i $N_t(P_i)$: the number of patents at P_i

The higher the index value, the more likely emerging technologies appear. The index value listed in Table 9 indicates that the highest values are observed in P2; during this period, technology has proliferated. However, note that certain technologies may emerge and disappear within a relatively short time. Thus, the keywords that only appeared in a single period were also regarded as unique keywords.

4.5.2. In-depth analysis of technological element

Time series analysis can be focused on a single keyword (technological element) rather than on a collection of keywords (technological field). Two technological elements were analyzed in depth: “interface,” a keyword observed in patent abstracts and images, and “wifi,” a keyword observed only in the patent images in the case study (Fig. 8).

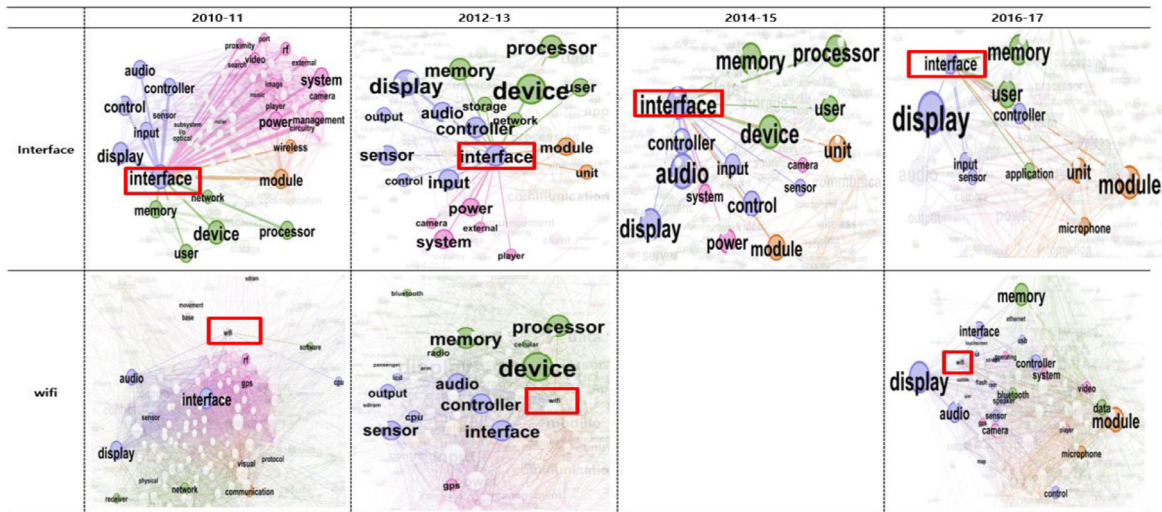


Fig. 8. In-depth analysis of “interface” and “wifi”.

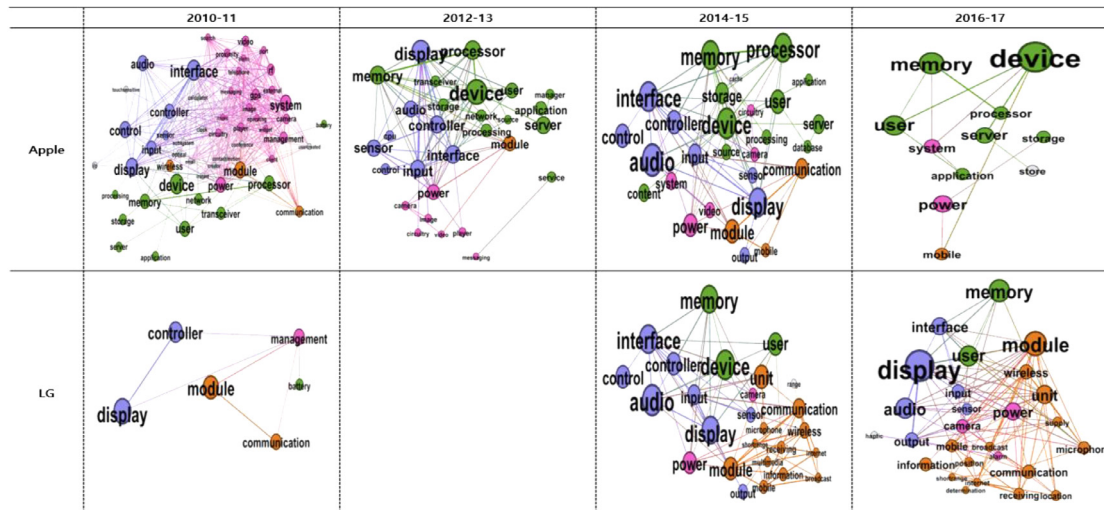


Fig. 9. Comparison of keyword networks: Apple and LG.

Fig. 8 provides information regarding the relationships of “interface” and “wifi” with other technological elements. First, “interface” was linked to various technologies in P2, particularly system-and-power-related technologies. However, its relationships with “audio,” “device,” and “display” were remarkable in P2; and it related to “audio,” “device,” and “processor” in P3 and to “user” in P4. The link between “interface” and “user” in P4 was distinct; the emphasis of “interface” moved from machines to users in P4. Moreover, a link between “interface” and “application” was established in P4; technologies that connect micro-earphones with mobile phones were actively developed. In contrast, wifi technologies were strongly linked with basic technologies, such as interface, audio, and display in P1, but were linked with the latest technologies that improved the usability of mobile earphones, including “bluetooth,” “speaker,” “Ethernet,” “LCD,” and “USB.”

4.5.3. Competitor technology analysis

The competitor trends in technological structure were analyzed using the patents by Apple and LG. Focusing on these two companies is worthwhile because they have the highest number of patents in the target technological field. The keywords and their positions in a network for each period are presented in Fig. 9.

Apple yielded more keywords than LG did because it had more patents: Apple had at least 30% of all keywords in each year. An interesting finding is that LG has strength in *module-related* technologies, whereas Apple has a competitive advantage in *device-related* technologies. Although these two firms are dominant in mobile earphone technologies, they have distinct differences in their targets for technology development. In particular, Apple had numerous keywords related to *system-and-power-related* technologies in P1 but

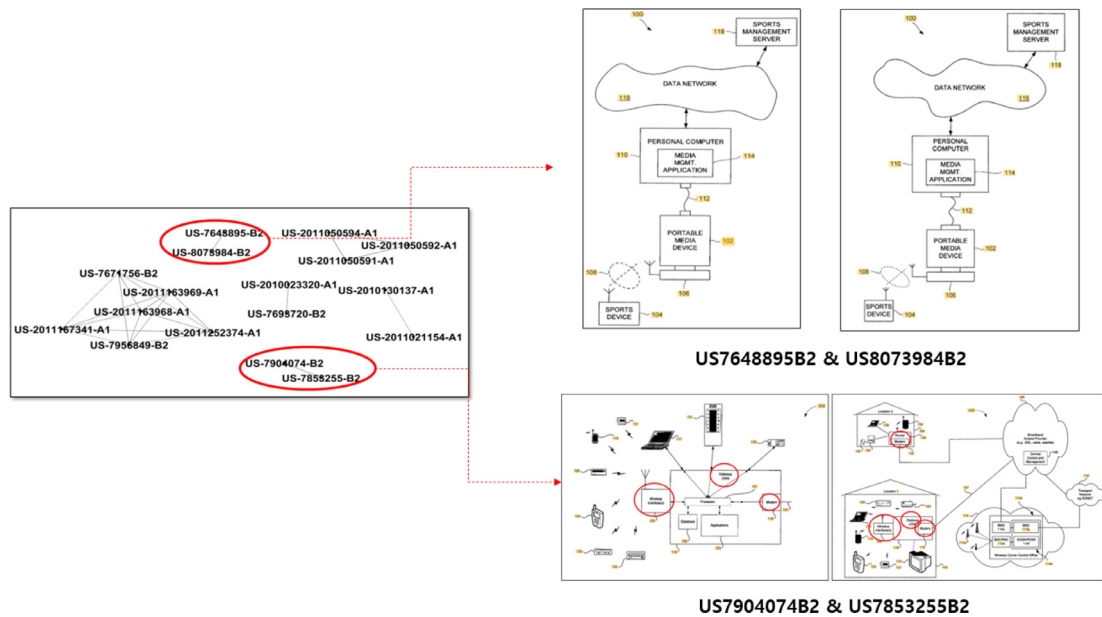


Fig. 10. Patent similarity analysis results.

had only a few in P4. Apple's number of keywords related to *interface-related* technologies also decreased in P4. In the case of LG, specialization increased toward *module-related* technologies.

4.5.4. Patent similarity analysis

Similar patents in terms of technological structure were identified using the Jaccard similarity index values. For this purpose, a network among patents using similarity values was constructed. Fig. 10 shows a network with a cut-off value of 1.0, limiting the analysis period to P1. In the figure, two patents US7643895B2 and US8073984B2, both belonging to Apple, had a relatively high similarity value, holding exactly the same figures. Similarly, two patents, US7904074B2 and US7853255B2, had a relatively high similarity value, sharing keywords, such as “wireless,” “interface,” “gateway,” and “modem.” The common features of the two images are as follows: (1) a rectangle with the keyword “modem” is positioned in a bigger rectangle with the keyword “gateway”; and (2) two keywords, “gateway” and “wireless,” are closely positioned. The proposed approach can help identify and compare similar patents in terms of images.

4.6. Discussion

The case study aims to provide four types of technology intelligence using block diagrams in patent drawings, which offer several issues for discussion. First, the proposed approach is one of the early attempts to determine the value of text in patent drawings. The text in images is expected to resolve the problem of text preprocessing. Thus, although only four types of technology intelligence are proposed, other useful applications of such texts are possible. For example, given that patent block diagrams are usually adopted to present technological elements, the text in these diagrams may be used to develop a technology tree that shows hierarchical relationships among technological elements. Considering that constructing a technology tree is one of the most difficult tasks in technology planning, a list of keywords from patent images, including their relationships, can assist a firm in developing a technology tree encompassing a comprehensive set of technologies.

Second, the scope of analysis in this study is limited to block diagrams, mostly used for describing the components of a patented invention and their relationships because heterogeneous information can be extracted by various types of images. However, its extension to other types of drawings, such as flowcharts, graphs, or chemical formulas, opens a wide range of possibilities to gain insight. For example, flowcharts consist of sequentially related keywords that provide the information about a workflow or process relative to the patented invention. The precedence relations can be identified using the flowcharts. Because keyword dependencies are indicated by arrows, information regarding technology relationships or procedures can be analyzed in more detail. Chemical formulas in patents are sometimes the most significant part of patent documents in some areas (e.g., pharmaceuticals), for which not only texts (atoms) but also their links (bonding) need to be analyzed accurately to understand the molecular structure exactly. Once the content of chemical formulas can be analyzed based on the texts as well as their relationships in patent images, the effort devoted to searching for and analyzing patents in relevant fields can be considerably reduced. Likewise, various other types of patent images can also be used for technology intelligence, thus they are worth analyzing. Furthermore, the analysis methods need to be customized to each type of image to obtain valuable technology intelligence. On the other hand, flowcharts are another type of image.

Third, the proposed approach can be elaborated by introducing newer techniques and other types of patent data. Image recognition techniques are undergoing continuous development. For example, the application of convolutional neural networks to image classification has yielded a high-performance level. Such advances in image analysis techniques have to be reflected in the extraction of technology intelligence from patent images. With the development of such techniques, the proposed approach can be elaborated. In addition, other up-to-date methods (e.g., word2vec) exist for creating a keyword matrix and building a network such as word2vec. However, instead of using keyword co-occurrence (semantic analysis), we used the topological structure of keywords in patent drawings, where the distance between keywords was considered. Nevertheless, combining the up-to-date methods for creating a keyword matrix and building a network is worth discussion, being complementary to the proposed methods. We also tried to identify the linkages (particularly, arrows) presented in block diagrams, which we failed to do.

Fourth, the proposed approach is based on single words, exploring the potential value of image data, although bigram can be used. Therefore, compared to the subject-action-object (SAO) or TRT approaches, the information provided from patent image (in particular, block-diagram) may be limited to the structure of product/service/technology elements. Nevertheless, the keywords in the patent image enable the extraction of purpose-based technology intelligence without noise (e.g., product/service components and their structure in case of block diagrams). The relationships are also based on the topological distances, unlike SAO and TRT that focus on semantic distances. Therefore, future research will address the possibilities of combining SAO and TRT with the keywords from patent images to investigate the relationships between keywords from multiple perspectives. Indeed, patent image analysis can be combined with other patent analyses of bibliometric and descriptive data (abstracts, claims, and descriptions) to offer a greater number of valuable insights. For example, in patent images, figures are provided to link the elements of drawings with the sentences in the main text. Accordingly, information regarding the elements of drawings can be extracted in more detail from the main text of patent documents when necessary. For example, SAO analysis, which analyzes functional relationships between the elements from patent texts, can offer valuable insights when used with the proposed approach, presenting the structural relationships (see [Appendix 5](#) for details). Hence, this can support the more thorough analysis of technological elements in patent drawings.

5. Conclusion

With the increasing value of intellectual property in businesses, the importance of patents is expected to become rapidly more prominent. This paper proposes a method that is focused on patent analysis studies using patent images—a technique that has not been previously presented. To achieve this goal, patent drawings in the form of block diagrams were collected, texts in the diagrams were extracted, keywords from the texts were selected, and relationships among keywords based on their locations in the diagram were identified. Keywords extracted from patent images (block diagrams) and those from patent abstracts were compared as well to understand the characteristics of keywords derived from images. The comparison results ultimately prove the value of patent images as a source of technology intelligence. Moreover, four types of approach to creating value from image analysis, particularly block diagrams, were established: (1) analysis of technology evolution, (2) in-depth investigation of technological elements, (3) comparative analysis with competitors, and (4) search for similar patents.

This study is anticipated to contribute to existing patent analysis research in two ways. First, this work is one of the few attempts to analyze patent images to derive technology intelligence. Although there are some studies on patent image analysis, most are limited to patent searches that are focused on resolving problems pertaining to the search for similar patents based on image data. The findings of this study may serve as a basis for further research on extracting valuable intelligence to aid in comprehending technology trends, supporting policies, and making strategic decisions. The proposed approach may be enriched by enhancing the image analysis methods. Furthermore, it may be combined with other patent analysis methods. Second, this study discovered the characteristics of texts in block diagrams. One of the key steps in applying text mining to patent data is the appropriate extraction and selection of keywords from numerous sentences. In block diagrams, only keywords that are regarded as important to the current technological elements are likely to be provided, aiding in the elimination of noisy information in technology analysis. The foregoing is also supported by an empirical analysis finding. Texts in block diagrams are closely related to technological elements.

Despite its valuable contributions, this study has three limitations. First, the relationships among keywords were determined based on the locations of block diagrams on the assumption that tightly connected keywords are closely located in the diagram. This assumption may be true for a collection of patent drawings; however, it may not always be valid for a single patent drawing. This limitation can be overcome by detecting the arrows linking the blocks in patent drawings or using the semantic information in the main text. Another approach can be to use the number of boxes between the two keywords. Accordingly, further study is required to consider the information about linkages explicitly mentioned in the patent images. Second, this study was limited to only the block diagrams of various types of patent drawings. A future study is necessary to customize the proposed approach to various other types of drawings, considering their characteristics. Third, this study focused on the image data for extracting technology intelligence in terms of structural relationships between keyword levels of technological elements. However, the image data can be combined with other types of data to offer the relationships from various other perspectives. For example, SAO analysis enables examination of the functional relationships between technological elements. Further in-depth research and analysis of the potential for integrating patent image analysis and patent semantic analysis is required. Finally, we adopted a simple single-word approach for two reasons; comparative analysis is necessary to understand the nature of keywords in patent drawings, and this is one of the earliest attempts to investigate the value of keywords in patent drawings. Nevertheless, further analysis is required to implement a more elaborated method that can help identify meaningful terms by using n-grams or pre-defined dictionaries on technical terms.

Appendix 1. Comparative analysis of frequency-based and TF-IDF-based approaches

Because a keyword selection strategy customized to the text in images is necessary, the two methods—frequency-based and TF-IDF-based—were implemented and compared in terms of their results. The analysis results are summarized in Table A1.

Table A2 summarizes the keywords commonly selected by both approaches (common keywords) and those selected using only one approach (unique keywords). Common keywords generally explain the technological content of inventions regardless of approaches, whereas the unique keywords chosen by TF-IDF include texts that are extremely specific to describe the inventions or those erroneously detected. Thus, we concluded that the use of a frequency-based approach could produce higher-quality keywords than TF-IDF could. Because the texts in block diagrams indicate the distinguishing features of inventions in the form of simple key phrases, the frequency-based approach seemed to yield a higher performance level than the TF-IDF-based approach did Fig. A1.

Appendix 2. Keywords extracted from patent images

To investigate the keywords extracted from patent images and compare them with patent abstracts, the keywords derived only from patent images were classified into six categories: activity-related, device-related, general, technology-related, person-related,

Table A1
Comparative analysis results I: Frequency and TF-IDF.

Sort by Term frequency				Sort by TF-IDF			
Term	TF	DF	TF-IDF	Term	TF	DF	TF-IDF
Module	425	88	4.78	Section	13	1	6.5
Device	279	145	1.91	Widget	63	11	5.25
Interface	182	105	1.72	Group	31	5	5.17
Display	174	123	1.40	Param	10	1	5
Unit	172	60	2.82	Codebook	10	1	5
User	162	88	1.82	Module	425	88	4.78
Audio	161	98	1.63	Track	13	2	4.33
Processor	153	92	1.65	Mean	17	3	4.25
Controller	152	79	1.90	Utility	12	2	4
System	146	77	1.87	Stimulation	8	1	4
Memory	145	96	1.49	Exh	8	1	4
Server	144	66	2.15	Mam	8	1	4
Data	135	63	2.11	Sharing	8	1	4
Input	120	75	1.58	Probe	8	1	4
Power	116	75	1.53	Rlc	7	1	3.5
Communication	96	61	1.55	Eqp	7	1	3.5
Control	96	70	1.35	Subband	7	1	3.5
Application	95	47	1.98	Solid	7	1	3.5
Wireless	91	52	1.72	Pdu	7	1	3.5
Storage	83	57	1.43	Sets	7	1	3.5

Table A2
Comparative analysis results II: Frequency and TF-IDF.

Unique keyword sorted by TF				Unique keyword sorted by TF-IDF			
Term	TF	DF	TF-IDF	Term	TF	DF	TF-IDF
Sgsn	12	7	1.5	Param	10	1	5
Shorrange	12	9	1.2	Stimulation	8	1	4
Gw	11	7	1.37	Mam	8	1	4
Comm	11	9	1.1	Exh	8	1	4
Touchsensitive	11	10	1	Eqp	7	1	3.5
Param	10	1	5	Definitions	6	1	3
Bs	10	4	2	Profiler	6	1	3
Mgmt	10	5	1.66	Pt	6	1	3
Pdsn	10	6	1.42	Ch	6	1	3
Pos	10	8	1.11	Whiteboard	6	1	3
Ram	9	7	1.12	Communic	5	1	2.5
Usercreated	9	8	1	Webmgt	5	1	2.5
Stimulation	8	1	4	Bee	5	1	2.5
Mam	8	1	4	Bs	10	4	2
Exh	8	1	4	Vf	6	2	2
Ma	8	4	1.6	Components	6	2	2
Controls	8	4	1.6	Mob	4	1	2
App	8	4	1.6	Cir	4	1	2
Structures	8	5	1.33	Commun	4	1	2
Msc	8	8	0.88	Cat	4	1	2

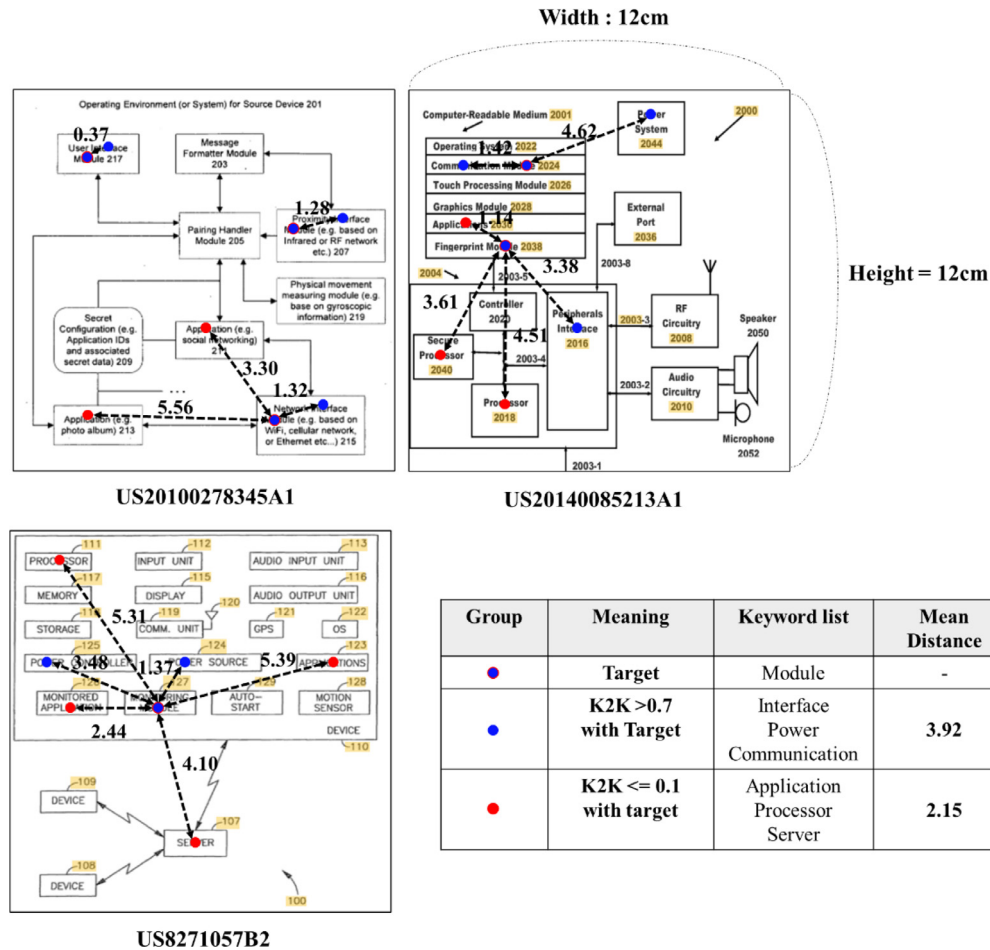


Fig. A1. The actual distance between keywords in patent images and K2K matrix values.

and others, as summarized in **Table A5**. More than 25% of the keywords were related to technology, followed by activity (21.5%) and device (21.5%). The keywords in the images indicate the technological elements, features of devices, and operational activities; hence, they are useful in analyzing technologies. Here, because the keywords in the others category are less informative than those in the other five categories are, they have been deleted from the final set of keywords for analysis.

Appendix 3. Comparative analysis of keywords from various parts of patent documents

To determine the importance of patent image data, we performed comparative analysis among the core keywords extracted from various sources of patent documents: title, abstract, claims, and image. These are the most commonly adopted sources for patent analysis. First, we applied keywords with “TF (term frequency) ≥ 4 ” to 635 patents, which produced 137 keywords for titles, 859 keywords for abstracts, 291 keywords for images, and 2584 keywords from claims. Then, among these keywords, we selected *unique keywords* for each source. We defined the keywords as those that do not exist in other sources. We also classified the unique keywords into two groups—*technical keywords* and *general keywords*—to investigate how many technical keywords each source generates. For this purpose, we used *Oxford 3000*, a general terminology dictionary, assuming that only general keywords are not presented in the dictionary. Finally, we controlled the total number of keywords in each source for more objective comparative analysis; the total number of unique keywords was set to 65–80 using the term frequency–inverse document frequency (TF–IDF) values. The number is not exactly the same in all four sources due to keywords with the same TF–IDF values. We designated these selected keywords as *core unique keywords*.

Table A3, **Table A4** summarizes the distribution of technical and general keywords among unique keywords and among core unique keywords. The results of the analysis indicate that patent images include a large number of unique keywords (28 out of 291 keywords). Furthermore, all of these are technical keywords in the case of images, which are important for understanding the technological characteristics of patented inventions. On the other hand, when TF–IDF is applied, a similar ratio of technical keywords is observed in all four sources. Nevertheless, the keywords in images are distinguished from others in that they tend to represent the technological elements in terms of product components. Thus, the texts in the images are still worth analysis.

Table A3

Classification of keywords observed only in patent images.

	activity	device	general	technology	person	others
counts	56	56	31	70	22	25
proportion	21.5%	21.5%	11.9%	26.9%	8.4%	9.6%
Keyword	search, capture, definition, volatile, shortrange, messaging, conversion, generation, reduction, short, high, contact, stimulation, description, mixer, cation, purchase ...	engine, encoder, decoder, cpu, generator, calculator, usb, accelerometer, ram, track, modulator, modem, lcd, ap, comm, selector, touchscreen, flash, browser, keypad ...	second, utility, dictionary, name, codebook, weather, calendar, first, facility, sport, central, solid, probe, right, parser, i/f, drive, ref, render, general, temperature ...	gps, graphic, baseband, subsystem, wifi, adc, web, codec, lan, stock, mgmt., param, tx, rx, email, dac, msc, portal ...	guest, usercreated, creator, merchant, estimator, face, bank, reader, operator, classifier, determiner, profiler, pm, issuer, companion, trainer ...	i/o, mam, exh, ma, a/d, ci, bt, im, pt, mme, al, proc, vre, ch, ir, drx, mc, bee, urunpy, mopule, ue, qutput, xcvr, imsi, ar

Table A4

Distribution of technical keywords in various sources of patent documents.

		Title	Abstract	Claims	Image
Unique keyword (TF)	Total	0/137	30/859	1708/2584	28/291
	Technical keyword samples	–	24 (80%) imc, mobile-phone, compositions, snet, aspects, pmi, disclosure, implementations, embodiments, pdu, ...	1243 (72.7%) chipset, bundle, resonator, satellite, intake, rbs, deployment, negotiate, handwrite, peds, ...	28 (100%) eqp, ram, contact/motion, cations, s103, adc, trans, stimulation, sgns/pdsn, comm, ...
	Common keyword samples	–	6 (20%) pot, variety, suit, goods, advantage, efficient	465 (27.2%) leave, shock, series, permit, decide, give, surround, subject, show, author, ...	–
Core unique keyword (TF-IDF)	Total	75	65	80	79
	Technical Keyword samples	40 (53.3%) feedback, rout, protocol, modes, creation, enhance, devices, interference, aggregation, playback, ...	35 (46.6%) beam, cochlear, microwave, earphone, deadline, theater, customers, dma, imc, composition, ...	62 (77.5%) episode, disfavor, outcomes, pdcp, audiometer, employees, poi, contention, enterprise, diagnosis, ...	41 (51.8%) server, dc/dc, stimulation, outlet, sgns/pdsn, wifi, eqp, mechanism, transducer, ...
	Common Keyword samples	35 (46.6%) position, research, quality, view, access, schedule, entertainment, combine, platform, effect, ...	30 (46.1%) pin, advertise, ticket, student, doctor, arrange, prompt, layer, pilot, song, ...	18 (22.5%) column, risk, employee, context, patient, pronunciation, victim, core, underwater, examination, ...	38 (48.1%) file, resource, rank, delay, zone, credit, learn, frame, level, contact, ...

Appendix 4. Validation of K2K matrix

The K2K matrix was validated by comparing the distance between keywords in the matrix and in the actual patent images. For this validation, the most frequently observed keyword “module” was selected to be compared with two groups of three randomly selected keywords; one is composed of the keywords closely related with “module” with the cell value greater than 0.7 (“interface”, “power”, and “communication”), whereas the other consists of the keywords less close to “module” with the cell value smaller than 0.1 (“application”, “processor”, and “server”). To consider the differences in image size, all the images were adjusted to the size of 12 cm by 12 cm. We also used the images having all the seven keywords, where the shortest distances between keywords in actual images were measured as shown in Fig. A2. In actual images, the keywords with relatively small distance (the degree of relationship greater than 0.7) are farther away from the target keyword than the keywords with relatively large distance (the degree of relationship smaller than 0.1) by 1.82 times on average. This indicates that the global K2K matrix can represent the distance between keywords well.

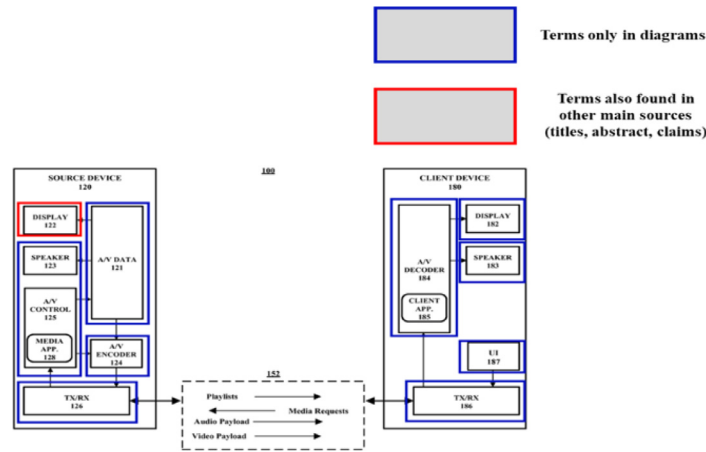


Fig. A2. Classification of common terms in patent block diagrams (US20130238702A1).

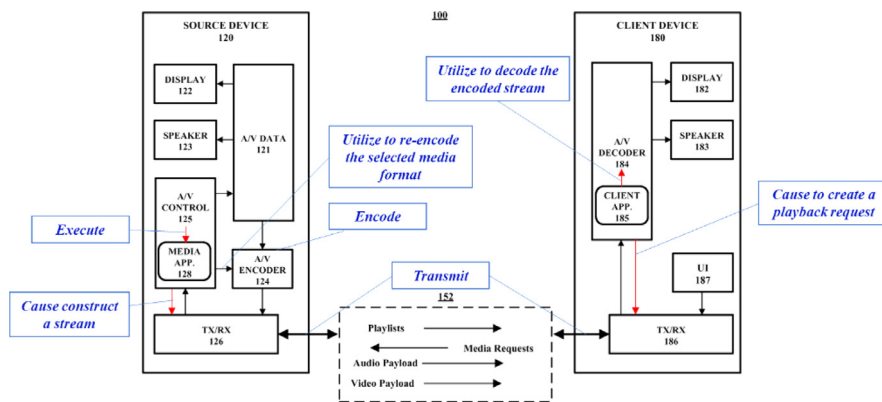


Fig. A3. Diagram relationship combined with SAO analysis (US20130238702A1).

Appendix 5. Investigation of potential value for combining SAO analysis with image analysis

The proposed approach can be combined with SAO analysis to create more value, with the two methods being complementary to each other. Image analysis enables us to identify technological elements relatively easily without noise and to investigate their structural relationships presented in block diagrams, while the SAO analysis enables an in-depth analysis of the functional relationships between the elements.

Focusing only on a single, randomly selected patent (US20130238702A1), we explored the potential value of integrating the image analysis with the SAO analysis. First, we examined whether the keywords presented in the block diagrams also appear in other parts of patent documents. The results of the analysis show that few of the keywords in the block diagrams are observed in patent titles, abstracts, and claims (see Fig. A2); the analysis reveals that the information (titles, abstracts, and claims) commonly used for patent analysis may not reflect the structural characteristics of the technology. Although all the keywords exist in the patent description, analyzing the full description for this purpose may be inefficient, suggesting that the texts in images have potential as an alternative approach.

Next, we investigated the potential of combining the information from SAO analysis with that from the image analysis. When restricting S and A to the keywords identifiable in the image analysis, eight relationships were available from the results of the SAO analysis. These results could deliver additional valuable information about the image analysis results (see Fig. A3). The findings indicate that not all relationships between technological elements presented in the block diagrams are extracted from the SAO analysis, but the functional relationships identified by SAO analysis are expected to add much value to the structural relationships identified through image analysis, signifying the potential of combining the two analysis methods. Future analysis will address this issue.

CRedit authorship contribution statement

Jeonghun Jee: Conceptualization, Methodology, Writing – original draft, Software. **Sanghyun Park:** Data curation, Software, Validation, Visualization. **Sungjoo Lee:** Validation, Investigation, Writing – review & editing.

References

- An, J., Kim, K., Mortara, L., & Lee, S. (2018). Deriving technology intelligence from patents: Preposition-based semantic analysis. *Journal of Informetrics*, 12(1), 217–236.
- An, X., Li, J., Xu, S., Chen, L., & Sun, W. (2021). An improved patent similarity measurement based on entities and semantic relations. *Journal of Informetrics*, 15(2), Article 101135.
- S. Chakrabarti, B. Dom, R. Agrawal, & P. Raghavan (1998). Scalable feature selection, classification and signature generation for organizing large text databases into hierarchical topic taxonomies, *The VLDB Journal*, 7(3), 163–178.
- Choi, J., & Hwang, Y. S. (2014). Patent keyword network analysis for improving technology development efficiency. *Technological Forecasting and Social Change*, 83, 170–182.
- Choi, S., Kim, H., Yoon, J., Kim, K., & Lee, J. Y. (2013). An SAO-based text-mining approach for technology roadmapping using patent information. *R&D Management*, 43(1), 52–74.
- Choi, S., Park, H., Kang, D., Lee, J. Y., & Kim, K. (2012). An SAO-based text mining approach to building a technology tree for technology planning. *Expert Systems With Applications*, 39(13), 11443–11455.
- Codina-Filbà, J., Bouayad-Agha, N., Burga, A., Casamayor, G., Mille, S., Müller, A., et al. (2017). Using genre-specific features for patent summaries. *Information Processing & Management*, 53(1), 151–174.
- Csurka, G. (2017). Document image classification, with a specific view on applications of patent images. In *Current challenges in patent information retrieval* (pp. 325–350). Berlin, Heidelberg: Springer.
- Daim, T. U., Rueda, G., Martin, H., & Gerdts, P. (2006). Forecasting emerging technologies: Use of bibliometrics and patent analysis. *Technology Forecasting and Social Change*, 73(8), 981–1012.
- Eakins, J. P. (2001). Trademark image retrieval. *Principles of visual information retrieval*. Berlin: Springer Verlag.
- Fall, C. J., Torcsvari, A., Benzineb, K., & Karetka, G. (2003). Automated categorization in the international patent classification. *ACM SIGIR Forum*, 37(1), 10–25.
- Hanbury, A., Bhatti, N., Lupu, M., & Mörzinger, R. (2011). Patent image retrieval: A survey. In *Proceedings of the 4th workshop on patent information retrieval* (pp. 3–8).
- Huang, M. H., Yang, H. W., & Chen, D. Z. (2015). Increasing science and technology linkage in fuel cells: A cross citation analysis of papers and patents. *Journal of Informetrics*, 9(2), 237–249.
- Kim, B., Gazzola, J., Lee, D., Kim, K., & Jeong, M. (2014). Inter-cluster connectivity analysis for technology opportunity discovery. *Scientometrics*, 98(3), 1811–1825.
- Kim, J. H., & Choi, K. S. (2007). Patent document categorization based on semantic structural information. *Information Processing & Management*, 43(5), 1200–1215.
- Kim, K., Park, K., & Lee, S. (2019). Investigating technology opportunities: The use of SAOX analysis. *Scientometrics*, 118, 45–70.
- Kim, Y. G., Suh, J. H., & Park, S. C. (2008). Visualization of patent analysis for emerging technology. *Expert Systems with Applications*, 34(3), 1804–1812.
- Lai, K. K., & Wu, S. J. (2005). Using the patent co-citation approach to establish a new patent classification system. *Information Processing & Management*, 41(2), 313–330.
- Lee, C., Kogler, D. F., & Lee, D. (2019). Capturing information on technology convergence, international collaboration, and knowledge flow from patent documents: A case of information and communication technology. *Information Processing & Management*, 56(4), 1576–1591.
- Lee, S., Yoon, B., & Park, Y. (2009). An approach to discovering new technology opportunities: Keyword-based patent map approach. *Technovation*, 29(6), 481–497.
- Lee, Y., Kim, S. Y., Song, I., Park, Y., & Shin, J. (2014). Technology opportunity identification customized to the technological capability of SMEs through two-stage patent analysis. *Scientometrics*, 100(1), 227–244.
- Li, R., Wang, L., & Hong, C. (2009). Extracting the significant-rare keywords for patent analysis. *Expert Systems with Applications*, 36(3), 5200–5204.
- Liang, Y., Tan, J., & Ma, J. (2018). Patent analysis with text mining for TRIZ. In *Proceedings of the 4th IEEE international conference on management of innovation and technology* (pp. 1147–1151).
- Noh, H., Jo, Y., & Lee, S. (2015). Keyword selection and processing strategy for applying text mining to patent analysis. *Expert Systems With Applications*, 42(9), 4348–4360.
- Pilkington, A. (2003). Technology commercialisation: Patent portfolio alignment and the fuel cell. In *Proceedings of the PICMET'03: Portland international conference on management of engineering and technology management for reshaping the world: 2003* (pp. 400–407).
- Pilkington, A., Lee, L. L., Chan, C. K., & Ramakrishna, S. (2009). Defining key inventors: A comparison of fuel cell and nanotechnology industries. *Technological Forecasting and Social Change*, 76(1), 118–127.
- Porter, A. L. (2005). QTIP: Quick technology intelligence processes. *Technological Forecasting and Social Change*, 72(9), 1070–1081.
- Porter, A. L., Watts, R. J., & Anderson, T. R. (2003). Mining PICMET: 1997–2003 papers help you track management of technology developments. In *Proceedings of the PICMET'03: Portland international conference on management of engineering and technology management for reshaping the world: 2003* (pp. 188–193).
- Satyawati, W., Pratama, M. O., Jannati, R., Muhammad, G., Fajar, B., Hamzah, H., et al. (2019). Citizen Id card detection using image processing and optical character recognition. *Journal of Physics: Conference Series*, 1235(1), Article 012049 IOP Publishing.
- Son, C., Suh, Y., Jeon, J., & Park, Y. (2012). Development of a GTM-based patent map for identifying patent vacuums. *Expert Systems with Applications*, 39(3), 2489–2500.
- Song, G., Huang, X., Cao, G., Liu, W., Zhang, J., & Yang, L. (2019). Enhanced deep feature representation for patent image classification. In *Proceedings of the volume 11069, 10th international conference on graphics and image processing (ICGIP 2018)*.
- Sporici, D., Cuşnir, E., & Boiangiu, C. A. (2020). Improving the accuracy of tesseract 4.0 OCR engine using convolution-based preprocessing. *Symmetry*, 12(5), 715.
- Stefanos, V., Anastasia, M., & Ioannis, K. (2012). Concept-based patent image retrieval. *World Patent Information*, 34(4), 292–303.
- Stefanos, V., Symeon, P., Anastasia, M., Panagiotis, S., Emanuelle, P., & Ioannis, K. (2010). Towards content-based patent image retrieval: A framework perspective. *World Patent Information*, 32(2), 94–106.
- Tiwari, A., & Bansal, V. (2004). PATSEEK: Content-based image retrieval system for patent database. In *Proceedings of the international conference on electronic business-04*. Tsinghua University.
- Tseng, Y. H., Lin, C. J., & Lin, Y. I. (2007a). Text mining techniques for patent analysis. *Information Processing & Management*, 43(5), 1216–1247.
- Vrochidis, S., Papadopoulos, S., Moutzidou, A., Sidiropoulos, P., Pianta, E., & Kompatsiaris, I. (2010). Towards content-based patent image retrieval: A framework perspective. *World Patent Information*, 32(2), 94–106.
- Wartburgh, I. V., Teichert, T., & Rost, K. (2005). Inventive progress measured by multi-stage patent citation analysis. *Research Policy*, 34(10), 1591–1607.
- Wu, J. K., Lam, B. M., Mehtre, Y. J., & Gao, A. (1996). Content-based retrieval for trademark registration. *Multimedia Tools and Applications*, 3, 245–267.
- Xie, Z., & Miyazaki, K. (2013). Evaluating the effectiveness of keyword search strategy for patent identification. *World Patent Information*, 35(1), 20–30.
- Yoon, B., Lee, S., & Lee, G. (2010). Development and application of a keyword-based knowledge map for effective R&D planning. *Scientometrics*, 85(3), 803–820.
- Yoon, B., & Park, Y. (2004). A text-mining-based patent network: Analytical tool for high-technology trend. *Journal of High Technology Management Research*, 15, 37–50.
- Yoon, J., Park, H., & Kim, K. (2013). Identifying technological competition trends for R&D planning using dynamic patent maps: SAO-based content analysis. *Scientometrics*, 94(1), 313–331.
- Zhang, L. (2011). Identifying key technologies in Saskatchewan, Canada: Evidence from patent information. *World Patent Information*, 33(4), 364–370.
- Tseng, Y. H., Wang, Y. M., Juang, D. W., & Lin, C. J. (2005). Text mining for patent map analysis, *Proceedings of IACIS Pacific 2005 Conference*, 1109–1116.